

# INTERNSHIP REPORT

## OF

# **C++ PROGRAMMING**

Submitted  
To

**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
B.Tech CSE



**BY**

Roll Number of Student: 25SCS1003000705

Name of Student: P VEDANSH

Batch: 2025–2029

Semester / Section: 3rd / 2CSE32

**2026–27**

## CANDIDATE'S DECLARATION

I, P VEDANSH, do hereby declare that the internship report titled “C++ Programming” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on the internship activities, projects, and documentation completed during my internship at Thiranex. All references and external materials used in preparing this report have been appropriately acknowledged. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement, and I shall be solely responsible for any copyright violation in this regard.

Signature: \_\_\_\_\_

Student Name: P VEDANSH

Roll No.: 25SCS1003000705

## ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout my internship in C++ Programming. The internship provided me with an opportunity to strengthen my programming fundamentals through practical, project-based tasks and to understand how software applications are designed, implemented, tested, and documented.

I am thankful to Thiranex for providing me with the opportunity to undertake the internship as an Intern – C++ Programming. I appreciate the project-based learning environment through which I worked on practical console applications involving file handling, Object-Oriented Programming, functions, arrays, loops, conditional logic, and input/output operations.

I would also like to express my gratitude to IILM University, Greater Noida, and the School of Computer Science and Engineering for their academic support and encouragement. Finally, I express my gratitude to my parents for their blessings and continued support.

Signature: \_\_\_\_\_

Student Name: P VEDANSH

Roll No.: 25SCS1003000705

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## INTERNSHIP COMPLETION CERTIFICATE



*Figure 3.1 Internship Completion Certificate issued by Thiranex*

The certificate confirms that P Vedansh successfully completed the internship in C++ Programming from 25 July 2026 to 24 August 2026. The verified certificate ID is THX-JUL2526-371.

## INTERNSHIP OFFER LETTER



Date: 25 July 2026

TO,

**P Vedansh**

Intern ID: THX-JUL2526-371

**Subject: Offer Letter for Internship in C++ Programming**

Dear **P Vedansh**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

|                   |                                 |
|-------------------|---------------------------------|
| Internship Role   | <b>Intern - C++ Programming</b> |
| Commencement Date | <b>25 Jul 2026</b>              |
| Completion Date   | <b>24 Aug 2026</b>              |
| Work Mode         | <b>Remote / Project-Based</b>   |

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.  
For verification, contact Thiranex Verification Cell.

[www.thiranex.in](http://www.thiranex.in)



Figure 3.2 Internship Offer Letter issued by Thiranex

The offer letter identifies the internship role as Intern – C++ Programming, with a commencement date of 25 July 2026, completion date of 24 August 2026, and work mode of Remote / Project-Based.

## 4. PROJECT DESCRIPTION

### 4.1 Introduction

This report presents the work completed during the C++ Programming internship at Thiranex from 25 July 2026 to 24 August 2026. The internship was conducted in a remote, project-based mode and focused on practical programming tasks. According to the project repository, the internship was designed around building console-based applications and applying core C++ programming concepts in progressively more application-oriented problems.

The internship provided practical exposure to functions, arrays, loops, conditional statements, structures, file handling, input/output, and Object-Oriented Programming. The completed work consisted of four projects: Student Management System, Bank Management System, Library Management System, and Tic Tac Toe. Together, these projects provided practice in data management, persistence, menu-driven program design, object-oriented modelling, searching, validation, and game logic.

### 4.2 Organization Profile

Thiranex was the organization providing the internship opportunity. The offer letter issued to P Vedansh describes the position as Intern – C++ Programming and specifies a Remote / Project-Based work mode. The internship was structured around practical projects under industry mentorship, with progress reviewed during the internship period.

The internship therefore functioned as a practical bridge between academic C++ programming concepts and their application in small software systems. The work was organized around clearly defined tasks, implementation, testing, documentation, and submission through the project repository.

### 4.3 Student and Internship Details

| Particular              | Details                       |
|-------------------------|-------------------------------|
| Student Name            | P Vedansh                     |
| Roll Number             | 25SCS1003000705               |
| Session                 | 2025–2029                     |
| Semester                | 3rd                           |
| Section                 | 2CSE32                        |
| Organization            | Thiranex                      |
| Internship Role         | Intern – C++ Programming      |
| Work Mode               | Remote / Project-Based        |
| Internship Period       | 25 July 2026 – 24 August 2026 |
| Verified Certificate ID | THX-JUL2526-371               |

#### 4.4 Problem Statement

The internship addressed the practical problem of translating C++ programming concepts into functional, user-oriented console applications. While individual programming constructs can be learned in isolation, effective software development requires combining them to manage data, implement operations, validate user input, maintain persistent information, and provide a clear interaction flow.

The four internship tasks addressed this problem through different application scenarios. Student Management System focused on maintaining student records; Bank Management System simulated basic account operations; Library Management System handled books and members; and Tic Tac Toe applied arrays, functions, conditions, and game logic to an interactive two-player application.

#### 4.5 Project Objectives

1. Strengthen practical programming skills using C++.
2. Apply functions, arrays, loops, conditional statements, and input/output in complete programs.
3. Implement persistent data storage using file handling.
4. Understand and apply basic Object-Oriented Programming using classes and objects.
5. Develop menu-driven console applications with clear user interaction.
6. Practice data searching, updating, deletion, and retrieval.
7. Implement validation and decision logic for application workflows.
8. Develop, test, debug, and organize project code using Visual Studio Code, GCC, and GitHub.
9. Gain experience in breaking application requirements into manageable modules and tasks.

#### 4.6 Scope of the Project

The scope of the internship covered the design and implementation of small-scale C++ console applications. The projects were educational and project-based rather than live production systems. The Student, Bank, and Library Management Systems demonstrate basic data-management workflows, while Tic Tac Toe demonstrates interactive game logic.

The repository also contains task-specific reports and presentations along with the source code. The completed projects can serve as a foundation for future

enhancement, such as stronger validation, richer user interfaces, database integration, exception handling, modular source organization, and deployment as graphical or web-based applications.

#### 4.7 Technologies and Tools Used

| Technology / Tool           | Purpose                                | Application in Internship                              |
|-----------------------------|----------------------------------------|--------------------------------------------------------|
| C++                         | Programming language                   | Implemented all four projects                          |
| GCC                         | Compiler                               | Compiled and tested C++ source files                   |
| Visual Studio Code          | Integrated development environment     | Code writing, editing, and debugging                   |
| GitHub                      | Version control and repository hosting | Organized and submitted internship projects            |
| File Handling               | Persistent storage                     | Stored student, bank, and library records              |
| Object-Oriented Programming | Software modelling                     | Classes and objects in Bank and Library systems        |
| Functions                   | Modular programming                    | Reusable operations across projects                    |
| Arrays                      | Data representation                    | Tic Tac Toe board and structured data handling         |
| Loops & Conditional Logic   | Control flow                           | Menus, validation, repeated operations, and game logic |

#### 4.8 System Architecture

The projects follow a simple console-application architecture. The user interacts with a menu or game interface; input is processed by application logic implemented through functions, structures, classes, arrays, and conditional statements; and data is displayed or persisted using file handling where required. This architecture is appropriate for the educational scope of the internship because it clearly separates user interaction, processing logic, and persistent data.

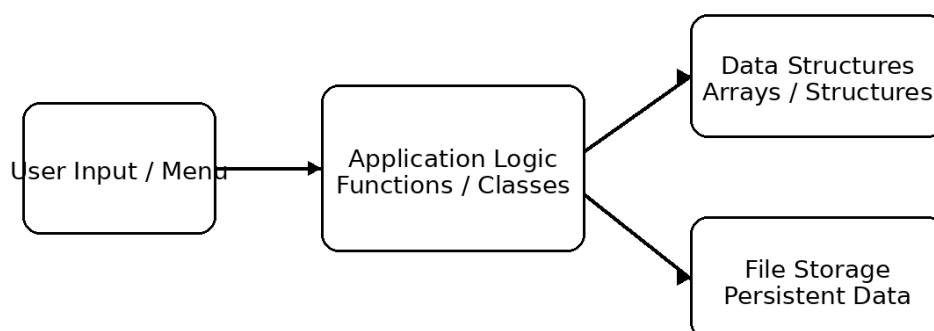


Figure 4.1 General architecture of the internship project workflow

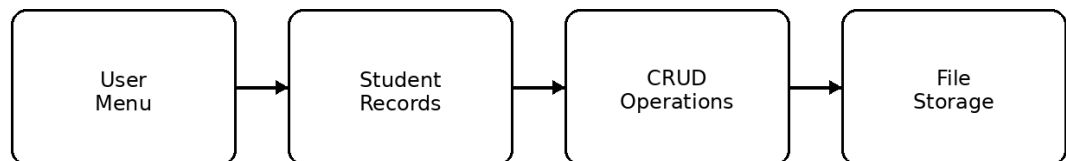
## 4.9 Methodology

The internship methodology was project-based. Each task began with understanding the required application behaviour, followed by program design, implementation in C++, compilation, testing, debugging, and repository organization. The projects increased the range of concepts being applied, from basic record management to object-oriented systems and interactive game logic.

| Task   | Project                   | Primary Methodology                                               |
|--------|---------------------------|-------------------------------------------------------------------|
| Task 1 | Student Management System | Menu-driven CRUD operations, structures, functions, file handling |
| Task 2 | Bank Management System    | Classes/objects, account operations, functions, file handling     |
| Task 3 | Library Management System | Classes/objects, book/member management, searching, file handling |
| Task 4 | Tic Tac Toe               | 3×3 array, two-player input, win/draw detection, replay logic     |

### 4.9.1 Student Management System

Figure 4.2 Student Management System workflow



The Student Management System is a console-based application for managing student records. Its core operations are adding, updating, deleting, and displaying student information. The project also uses file handling so that records can be stored persistently rather than existing only during program execution.

#### Main Features

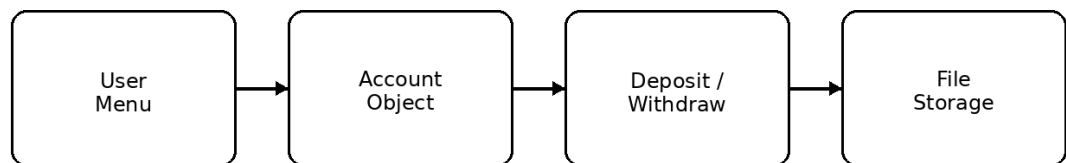
- Add student records
- Update student information
- Delete student records
- Display student records
- Store records using file handling
- Menu-driven interface

#### Concepts Used

- C++ basics
- Structures
- Functions
- File handling
- Input/output
- Conditional statements
- Loops

#### 4.9.2 Bank Management System

*Figure 4.3 Bank Management System workflow*



The Bank Management System is a console-based application that simulates basic banking operations. It models account-related information and provides operations such as account creation, deposit, withdrawal, balance checking, and displaying account details. The repository identifies Object-Oriented Programming, classes, objects, functions, and file handling as key concepts used.

##### **Main Features**

- Create a bank account
- Deposit money
- Withdraw money
- Check account balance
- Display account details
- Store account information using files

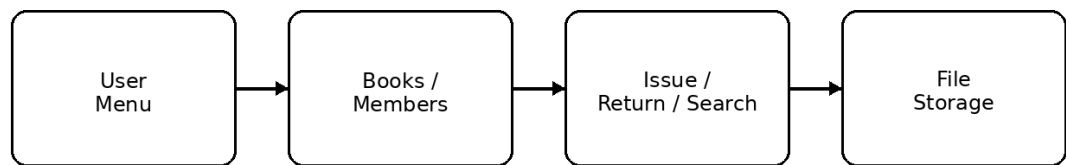
##### **Concepts Used**

- C++
- Object-Oriented Programming
- Classes and objects

- Functions
- File handling
- Conditional statements
- Loops

### 4.9.3 Library Management System

*Figure 4.4 Library Management System workflow*



The Library Management System is a console-based application for managing books and library members. It supports adding books and members, issuing and returning books, searching for books, and displaying stored information. The project combines object-oriented concepts with searching, control flow, and file handling.

#### **Main Features**

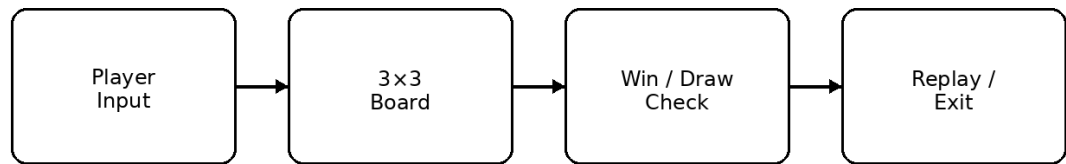
- Add books
- Add members
- Issue books
- Return books
- Search for books
- Display books and members

#### **Concepts Used**

- C++
- Classes and objects
- Functions
- File handling
- Loops
- Conditional statements
- Searching

#### 4.9.4 Tic Tac Toe

Figure 4.5 Tic Tac Toe game workflow



Tic Tac Toe is a console-based two-player game developed in C++. The game uses a 3×3 board and alternates between Player X and Player O. The application dynamically displays the board, detects winning and draw conditions, and provides a replay option. The project demonstrates how arrays, loops, functions, conditional logic, and input handling can be combined to implement an interactive program.

##### **Main Features**

- 3×3 game board
- Two-player gameplay
- Player X and Player O
- Dynamic board display
- Win and draw detection
- Replay option

##### **Concepts Used**

- Arrays
- Loops
- Conditional statements
- Functions
- Game logic
- Input handling

#### 4.10 Testing and Debugging Approach

Testing during the internship focused on executing the programs through their available menu or gameplay flows and checking whether each required operation produced the expected result. Debugging involved identifying compilation errors,

logical errors, incorrect input handling, and issues in data persistence, followed by correction and retesting.

For management systems, representative operations include creating records, modifying or deleting records, displaying information, and checking whether stored data can be retrieved. For Tic Tac Toe, representative cases include valid moves, alternating turns, winning combinations, draw conditions, and replay behaviour. This approach helped verify the main functional requirements of each project.

#### **4.11 Skills and Learning Outcomes**

10. Developed practical experience in writing complete console-based C++ applications.
11. Improved understanding of functions and modular program design.
12. Applied arrays, structures, loops, conditional statements, and input/output in practical scenarios.
13. Gained hands-on experience with file input/output and persistent data storage.
14. Applied Object-Oriented Programming concepts through classes and objects.
15. Practiced menu-driven application design and user interaction.
16. Implemented searching and record-management operations.
17. Improved testing and debugging practices for C++ programs.
18. Used GitHub to organize, maintain, and submit internship project work.
19. Learned to convert a written problem statement into an implementable software solution.

#### **4.12 Expected Outcomes**

The expected outcome of the internship was practical competence in C++ programming and the ability to implement small applications independently. The repository demonstrates these outcomes through four completed projects covering record management, banking operations, library operations, and game development. The internship also resulted in improved familiarity with source-code organization, file handling, OOP, debugging, and GitHub-based project submission.

#### **4.13 Repository and Project Documentation**

The project repository contains the four task folders, C++ source files, task-specific reports, task-specific presentations, the internship completion certificate, and the main README file. The repository therefore serves as the primary technical record of the work completed during the internship.

GitHub Repository:  
[https://github.com/pvedansh1110/2025-29\\_P\\_Vedansh\\_25SCS1003000705\\_3rd\\_2CSE32](https://github.com/pvedansh1110/2025-29_P_Vedansh_25SCS1003000705_3rd_2CSE32)

#### **4.14 Certificates of Completion and Communication Proof**

The internship completion certificate and offer letter are attached in Chapter 3. The completion certificate verifies successful completion of the C++ Programming internship from 25 July 2026 to 24 August 2026 and carries the verified certificate ID THX-JUL2526-371. The offer letter confirms the internship role, dates, and Remote / Project-Based work mode.

#### **5. BIBLIOGRAPHY / REFERENCES**

20. Thiranex, “Offer Letter for Internship in C++ Programming,” dated 25 July 2026, Intern ID: THX-JUL2526-371.
21. Thiranex, “Certificate of Achievement,” Internship in C++ Programming, 25 July 2026 – 24 August 2026, Certificate ID: THX-JUL2526-371.
22. P Vedansh, “C++ Programming Internship” GitHub repository, containing internship source code, task reports, presentations, README documentation, and certificate.
23. ISO / academic formatting requirements as specified in the IILM University School of Computer Science and Engineering Internship Report Format, 2026–27.